



CHRIS MACE

ClimateWins – Interim report

November 2025



OBJECTIVE

Use machine learning to help predict the consequences of climate change while working as a data analyst at a European nonprofit organization



CLIENT

The client, ClimateWins, wants to know:

- ▶ Can we apply machine learning to weather data?
- ▶ Can we use machine learning to analyse historic maximums and minimums of temperature?
- ▶ Can we use machine learning to predict favourable weather conditions on certain days?

All while considering ethical concerns



HYPOTHESES

- ▶ Machine learning can analyse historic and current data to predict how mean temperature will change in future years
- ▶ Machine learning can predict if it is favourable to go out on a certain day
- ▶ Machine learning can predict the risk of a dangerous condition such as high wind speed



DATA

- ▶ Data collected by ECA&D – European Climate Assessment & Data Set Project
- ▶ Collects and validates daily climate data from meteorological stations to create datasets used for research, assessments, and understanding climate trends and impacts
- ▶ Collected from 1960 to 2022 across 18 weather stations in mainland Europe
- ▶ Main data set includes variables such as wind speed, mean temperature and sunshine for each day. Pleasant weather data set is a yes/no format on if it is good to go out



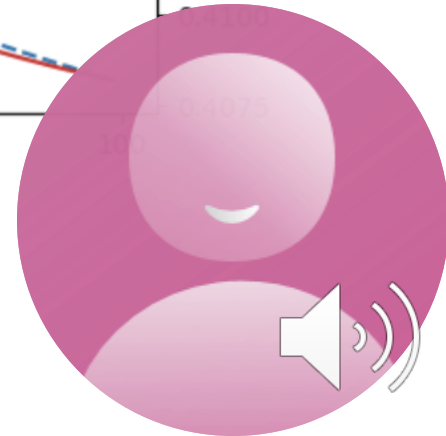
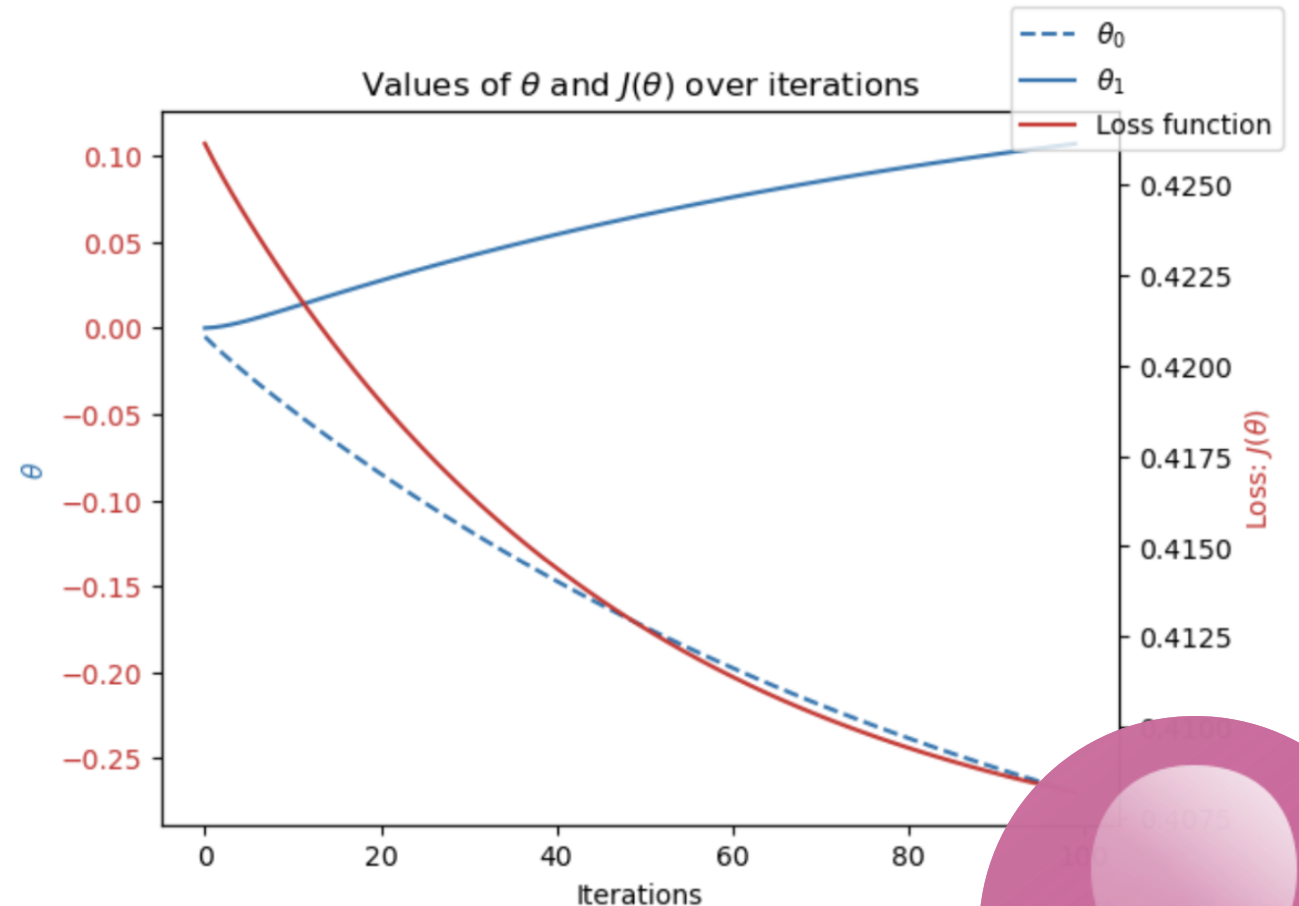
BIAS AND ACCURACY

- ▶ Location data purely from mainland Europe (selection bias)
- ▶ Vested interest of the provider potentially to highlight extremes; human may focus on say the change in temperature over other weather conditions such as windspeed or precipitation (interaction bias)
- ▶ Human error in data collection and preparation
- ▶ Accuracy – main weather data set seems complete with records for all 18 stations and their specific conditions, however pleasant weather dataset only for 15, and as we'll see – one is not useable!



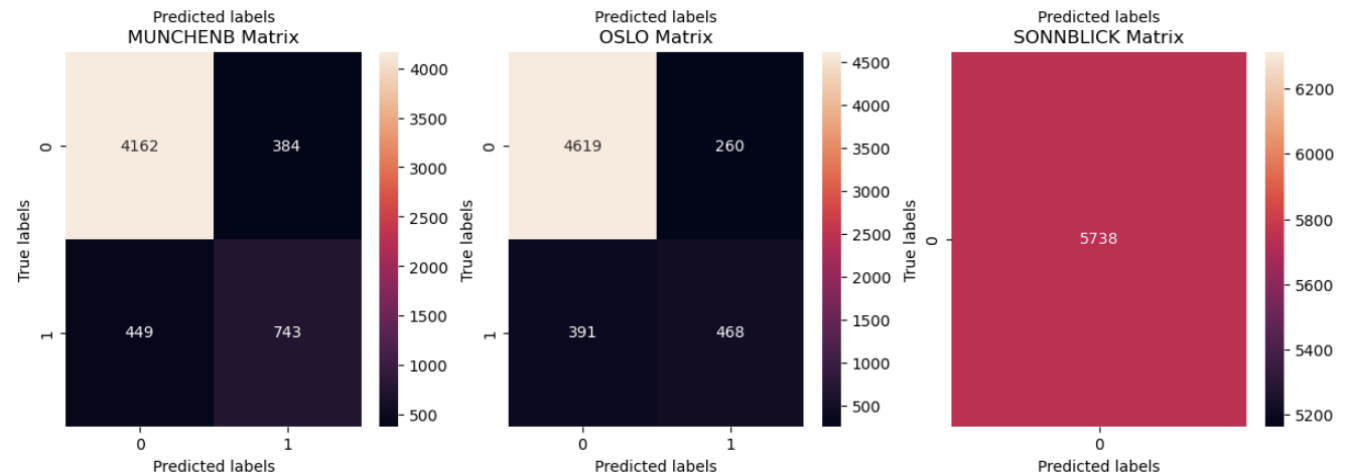
OPTIMISATION

- ▶ Optimisation, here using the gradient descent method, is used to find the best model parameters.
- ▶ It enabled us to minimize the the loss function while adjusting parameters such as iterations and step size, to ultimately improve model performance
- ▶ Best results using scaled data and this will apply to all models



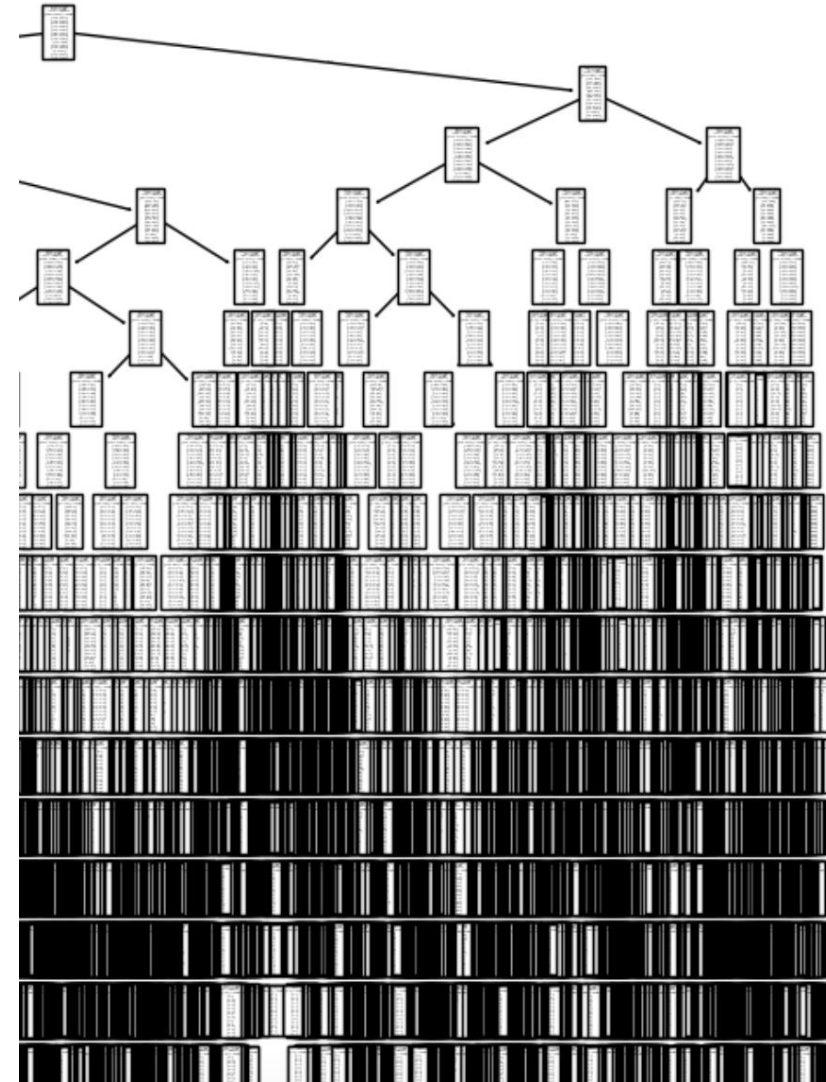
#1 K-NEAREST NEIGHBOUR (KNN)

- ▶ This algorithm is mainly used for classification, and works by measuring proximity between data points and then groups the closest into neighbours
- ▶ High accuracy – 85-90%
- ▶ We can already see issue with Sonnblick data set only contains 0s!



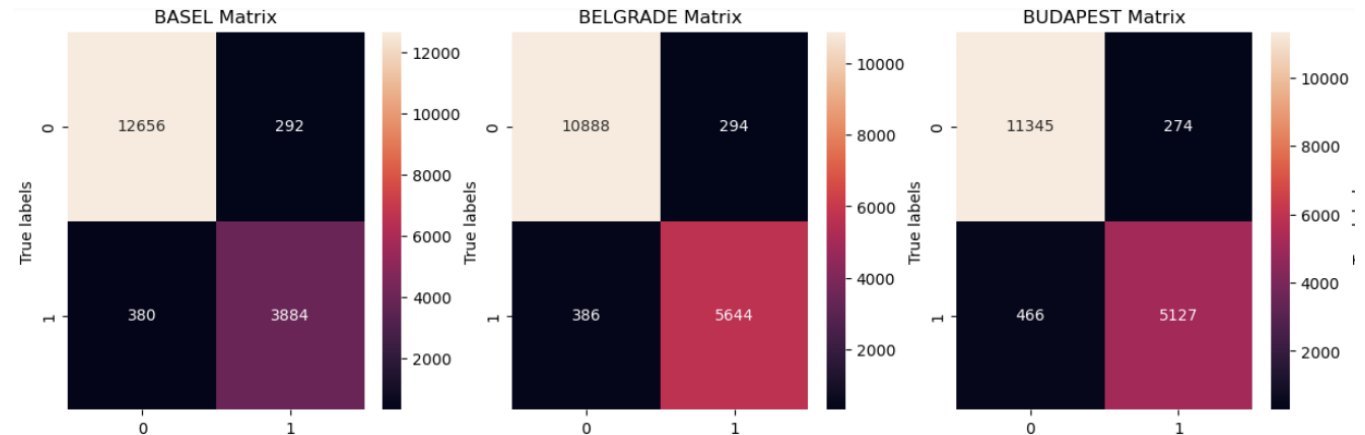
#2 DECISION TREE

- ▶ This algorithm has a flowchart-like structure used to split data and predict a result, with each node representing a decision point
- ▶ Difficult to visualise answers, long processing time, low accuracy of around 40%
- ▶ Pruning required as low accuracy likely from asking too many questions, making it prone to overfitting



#3 ARTIFICIAL NEURAL NETWORK (ANN)

- ▶ Computational algorithm using interconnected nodes (neurons) organized in layers, which process input data and generate an output, and learns by adjusting the strength of these connections (weights) to improve its performance
- ▶ I used various levels of parameters (nodes, iterations, and tolerance) in supervising this model
Training data maxed out at 72.5%
- ▶ Potential to improve and this flexibility required for our hypotheses



PREFERRED TYPE



While KNNs present higher accuracy, this is limited to simple classification. ANNs will be more useful over time as will scale better and learn complex weather data



ANNs can handle more variables, and we shall need to factor in all conditions when determining if the weather is suitable or potentially dangerous, and support or reject our hypotheses



SUMMARY

Hypotheses to be analysed using ANNs:

- ▶ Machine learning can analyse historic and current data to predict how the mean temperature will change in future years
- ▶ Machine learning can predict if it is favourable to go out on a certain day
- ▶ Machine learning can predict the risk of a dangerous condition such as high wind speed

Next steps and future analysis:

- ▶ Contact data collector to review how the data was prepared and minimise potential bias, especially for Sonnblick!
- ▶ Investigate advanced methods using ANNs to answer hypotheses



THANK
YOU!

QUESTIONS?

CHRIS MACE

CHRISMACE888@GMAIL.COM

